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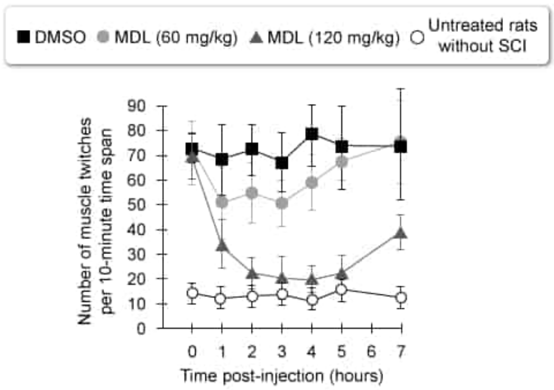

Figure 1 Muscle twitches in rats with SCI treated with MDL or DMSO and in untreated rats without SCI. Error bars represent 95% confidence intervals.

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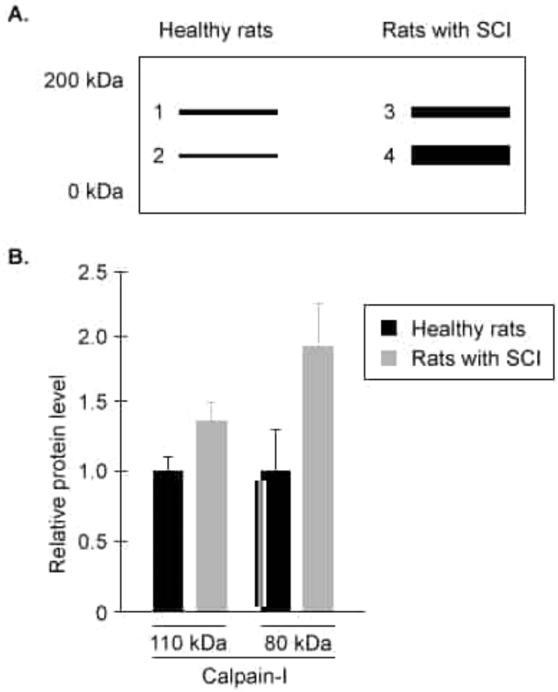


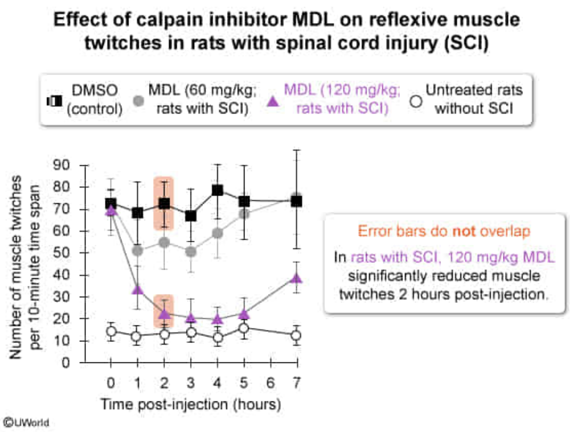
Figure 2 Calpain-I western blot (A) and quantified calpain-I expression levels (B) in each group

Which of the following statements best explains the data shown in Figure 1?

- A. Reflex arc overstimulation in muscle fibers was significantly reduced following treatment of rats with 60 mg/kg MDL. [2%]
- B. Treatment with 120 mg/kg MDL significantly reduced reflexive muscle twitches 2 hours after treatment. [90%]
- C. Reflex arc activity differed significantly among all post-injection time points in rats treated with 60 mg/kg MDL. [3%]
- D. Treatment with 120 mg/kg MDL was sufficient to permanently recover normal reflex arc activity in rats with SCI. [2%]

Correct
90% answered correctly

Explanation:



微信Mai_ange

The spinal cord of the central nervous system (CNS) plays a role in controlling reflexes (involuntary responses to stimuli). Some reflexes, called supraspinal reflexes, are also regulated by the brain or brainstem; however, many reflexes do *not* require brain input and are regulated solely by spinal cord neurons. Reflex arcs occur as follows:

1. **Sensory receptors** (specialized nerve endings) detect a stimulus (eg, chemical, pressure).

2. The stimulus activates the sensory neuron, which sends an afferent signal (ie, electrical impulse) *toward* spinal cord **integration centers** (regions that process sensory and CNS input).

3. Within the integration center, the signal may be passed directly to an **effector neuron** or to an **interneuron** that relays the signal to the effector neuron. For supraspinal reflexes, interneurons can also relay sensory signals to the brain and integrate brain signals to regulate reflex responses.

4. The activated effector neuron generates an **efferent** signal that travels from the spinal cord to the target tissue (eg, muscle), which carries out the appropriate response.

In the passage, researchers hypothesize that calpains (ie, proteases) are involved in SCI-induced **hyperreflexia** (ie, overactivity of reflex arcs causing muscle twitches). To test this, researchers injected rats expressing SCI with either a dose of the calpain inhibitor MDL (60 mg/kg or 120 mg/kg) or a control solution (DMSO). During each hour post-injection, researchers determined the average number of muscle twitches in each group occurring within a 10-minute time span.

Figure 1 shows that **2 hours after 120 mg/kg MDL treatment**, muscle twitching in SCI rats was **significantly reduced** compared to twitching in SCI rats treated with the DMSO control (ie, the confidence interval error bars do *not* overlap). However, 7 hours after this MDL injection, muscle twitches *increased* again (ie, to numbers significantly higher than those in rats without SCI), indicating that 120 mg/kg MDL was *not* sufficient to *permanently* recover normal reflex arc activity **(Choice D)**.

(Choices A and C) Because the error bars for SCI rats treated with 60 mg/kg MDL and those treated with DMSO overlap, 60 mg/kg MDL did *not* significantly reduce reflex arc overstimulation. In addition, for treatment with 60 mg/kg MDL, the error bars among all post-injection time points overlap, indicating that reflex arc activity did *not* differ significantly over time in rats receiving this treatment.

Educational objective:
In a reflex arc, a sensory neuron sends a signal to the spinal cord that is then sent (either directly or via an interneuron) to an effector neuron, which stimulates the desired response from the target tissue.

Research has indicated that spinal cord injury (SCI) in mammals leads to increased flow of sodium (Na^+) and calcium (Ca^{2+}) ions into affected motor neurons. In addition, activity of the KCC2 cotransporter protein, which normally functions to maintain low intracellular chloride ion (Cl^-) concentrations via active transport, has been found to be downregulated in motor neurons following SCI.

SCI-induced dysregulation of motor neurons can cause hyperreflexia, or overactivity of certain reflex arcs, leading to muscle twitches. Furthermore, some SCIs affecting upper spinal regions can prevent individuals from effectively regulating stress responses, including stress-induced blood pressure increases.

Calpains are proteases hypothesized to play a role in motor neuron dysfunction following SCI. To test this, researchers injected groups of rats all expressing the same type of SCI with either a dose of the calpain inhibitor MDL (60 mg/kg or 120 mg/kg) or a control solution not affecting calpain (DMSO). After the injection, researchers measured the average number of muscle twitches for each group within the first ten minutes of each hour. These values were compared to the mean number of muscle twitches occurring in untreated rats without SCI (Figure 1).

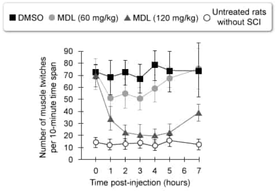


Figure 1 Muscle twitches in rats with SCI treated with MDL or DMSO and in untreated rats without SCI. Error bars represent 95% confidence intervals.

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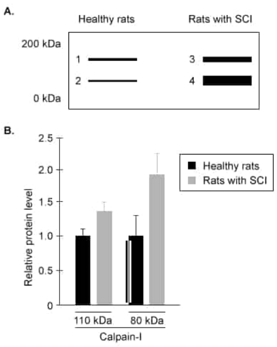


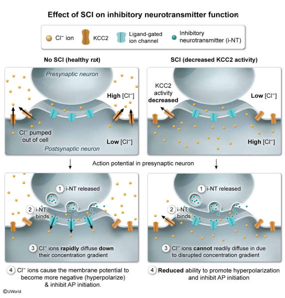
Figure 2 Calpain-I western blot (A) and quantified calpain-I expression levels (B) in each group

A neurotransmitter identified in the spinal cord of rats normally functions to bind ligand-gated Cl^- channels on motor neurons, facilitating the diffusion of Cl^- into the neurons and altering neuron membrane potential. Given this information, rats with downregulated KCC2 function in motor neurons due to SCI would most likely exhibit:

- A. abnormally rapid active transport of Cl^- out of motor neurons upon binding the neurotransmitter. [7%]
- B. abnormally rapid diffusion of Cl^- into these motor neurons following neurotransmitter release. [30%]
- C. reduced ability of the neurotransmitter to promote depolarization of cell membranes in these motor neurons. [28%]
- ✔ D. reduced ability of the neurotransmitter to promote hyperpolarization of cell membranes in these motor neurons. [32%]

Correct
32% answered correctly

Explanation:



A synapse (ie, a chemical or electrical junction) allows communication between the axon terminal of a **presynaptic neuron** and the dendrites of a **postsynaptic neuron**. In a presynaptic neuron, an action potential (AP) travels down the axon until reaching the axon terminals.

At chemical synapses, the AP causes the presynaptic neuron to secrete neurotransmitters into the synaptic cleft, which is the space between the presynaptic axon terminal and postsynaptic dendrite. The neurotransmitters bind receptors located on the postsynaptic neuron membrane, causing the opening of postsynaptic neuron ion channels. These ion channels facilitate the movement of charged ions into or out of the postsynaptic neuron, which alters the neuron's resting membrane potential (approximately −70 mV).

Synapses can also be excitatory (promoting AP initiation) or inhibitory (inhibiting AP initiation). In excitatory synapses, if the membrane potential of the postsynaptic neuron becomes more **positive** (ie, **depolarizes**) and exceeds a certain threshold (approximately −55 mV), an AP is initiated. However, in inhibitory synapses, neurotransmitters affect the postsynaptic neuron by causing either an influx of negative ions (Cl^-) or efflux of positive ions (K^+). This movement causes the cell's membrane potential to become *more* negative (ie, **hyperpolarize**), inhibiting AP initiation.

In this question, a spinal cord neurotransmitter in rats binds ligand-gated ion channels on motor neurons, allowing Cl^- diffusion into motor neurons. Accordingly, this neurotransmitter is **inhibitory**, as it would normally cause motor neuron membrane potential to become *more* negative, promoting **hyperpolarization** and inhibiting APs.

The passage states that the cotransporter KCC2 functions to maintain *low* intracellular Cl^- concentrations via active transport and that KCC2 activity is *downregulated* in motor neurons following SCI. The Cl^- concentration gradient established by KCC2 in motor neurons facilitates the flow of Cl^- into motor neurons bound by the inhibitory neurotransmitter, as Cl^- can diffuse from *higher* extracellular concentrations to *lower* intracellular concentrations.

In rats with SCI, *reduced* KCC2 activity would cause motor neurons to exhibit Cl^- concentrations *higher* inside the cell and *lower* outside the cell than in wild-type rats due to *decreased* active transport of Cl^- to the extracellular environment **(Choice A)**. This change would serve to decrease the concentration gradient for Cl^-, causing decreased (*not* abnormally rapid) diffusion of Cl^- into motor neurons via ion channels bound by this neurotransmitter **(Choice B)**. This neurotransmitter would display **reduced ability** to promote **hyperpolarization** of the membrane, and consequently would generally not be involved in promoting membrane depolarization **(Choice C)**.

Educational objective:
Neurotransmitter binding may cause neuron depolarization or hyperpolarization. Depolarization of a neuron cell membrane (ie, more positive membrane potential) promotes action potential initiation whereas hyperpolarization (ie, more negative membrane potential) inhibits action potential initiation.

Time spent:0 sec
Subject:
Biology

QID:403365
System:
Endocrine and Nervous Systems

Research has indicated that spinal cord injury (SCI) in mammals leads to increased flow of sodium (Na^+) and calcium (Ca^{2+}) ions into affected motor neurons. In addition, activity of the KCC2 cotransporter protein, which normally functions to maintain low intracellular chloride ion (Cl^-) concentrations via active transport, has been found to be downregulated in motor neurons following SCI.

SCI-induced dysregulation of motor neurons can cause hyperreflexia, or overactivity of certain reflex arcs, leading to muscle twitches. Furthermore, some SCIs affecting upper spinal regions can prevent individuals from effectively regulating stress responses, including stress-induced blood pressure increases.

Calpains are proteases hypothesized to play a role in motor neuron dysfunction following SCI. To test this, researchers injected groups of rats all expressing the same type of SCI with either a dose of the calpain inhibitor MDL (60 mg/kg or 120 mg/kg) or a control solution not affecting calpain (DMSO). After the injection, researchers measured the average number of muscle twitches for each group within the first ten minutes of each hour. These values were compared to the mean number of muscle twitches occurring in untreated rats without SCI (Figure 1).

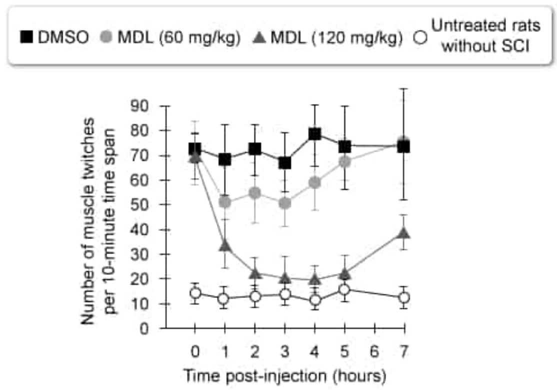


Figure 1 Muscle twitches in rats with SCI treated with MDL or DMSO and in untreated rats without SCI. Error bars represent 95% confidence intervals.

A specific calpain protein (calpain-I) exists in two forms, a larger (110 kDa) inactive form and a smaller (80 kDa) active form. Western blotting was used to test expression of calpain-I in spinal cord segments from two age-matched rat groups, a healthy control group that was untreated and a group in which SCI was surgically induced at a specific spinal cord location. The data were used to quantify relative calpain-I expression levels (Figure 2).

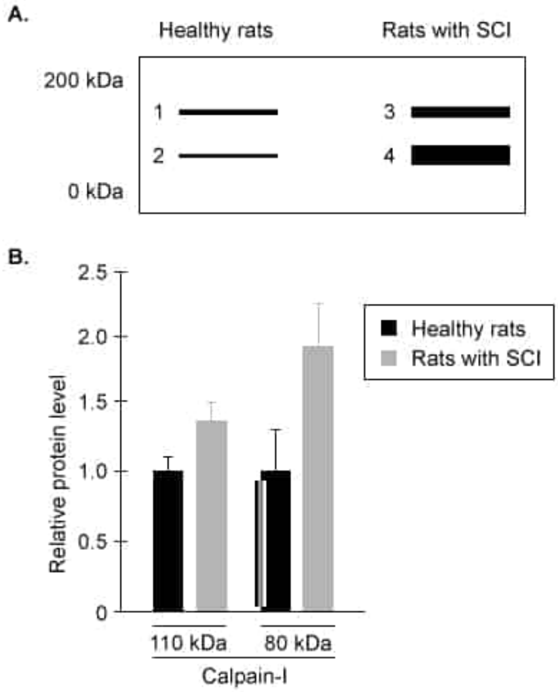


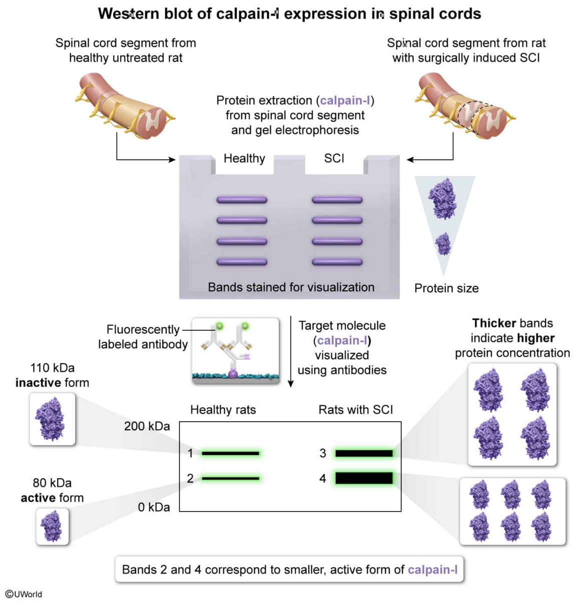
Figure 2 Calpain-I western blot (A) and quantified calpain-I expression levels (B) in each group

Which of the bands shown in the western blot in Figure 2 represent active calpain-I proteases?

- A. Band 4 only [9%]
- B. Bands 3 and 4 only [16%]
- C. Bands 2 and 4 only [66%]
- D. Bands 1 and 3 only [7%]

Correct
66% answered correctly

Explanation:



[Western blotting](#) is a method that can be used to detect the expression levels of specific proteins in a sample (eg, tissue, cell culture sample). During western blotting, the following steps are taken:

1. Proteins are extracted (eg, from tissue such as a spinal cord segment)
2. Gel electrophoresis is performed on the proteins to separate them by size. **Larger proteins will migrate a shorter distance through the gel than smaller proteins**.
3. Proteins are transferred from the gel to a protein-binding membrane (eg, nitrocellulose), where they are immobilized.
4. Protein-rich mixtures are used to block portions of the membrane to which proteins were *not* transferred. This prevents antibodies (ie, another type of protein) from binding nonspecifically to the membrane in steps 5 and 6.
5. The membrane is incubated with primary antibodies that bind specific proteins.
6. Secondary antibodies, which include fluorescent labels, are applied and bind primary antibodies.
7. Antibodies bound to the proteins are detected via fluorescence, allowing specific protein bands to be visualized. Thicker bands indicate that the protein is more highly expressed.

In the passage, western blotting was used to test expression of the protease calpain-I in spinal cord segments from a healthy control group of rats that was untreated and a group of rats in which SCI was surgically induced. The passage states that calpain-I exists in two forms, a larger (110 kDa) inactive form and a smaller (80 kDa) active form.

Therefore, because **active** calpain-I is **smaller in size**, these forms of the protein likely traveled *farther* through the gel than the larger inactive forms during the gel electrophoresis. Figure 2A shows that the smaller bands representing active calpain-I occur in both healthy rats and in the rats with SCI. Specifically, these 80 kDa proteins are represented by **bands 2 and 4** in the western blot results.

(Choices A, B, and D) These options represent *incorrect* combinations of bands on the western blot results.

Educational objective:
Western blotting can be used to detect the expression levels of proteins extracted from a tissue or cell sample. During western blotting, proteins separated via gel electrophoresis are transferred to a membrane incubated with fluorescently labeled antibodies, which allows specific protein bands to be visualized.

Time spent:0 sec	**QID**:403366
Subject: Biology	**System:** Endocrine and Nervous Systems

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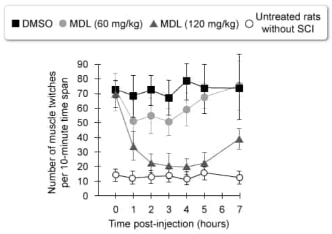

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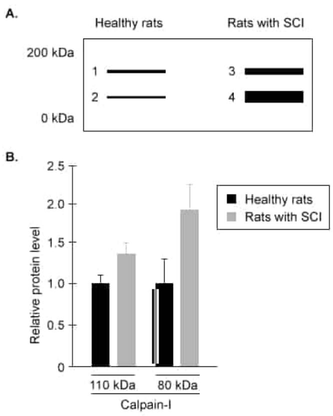


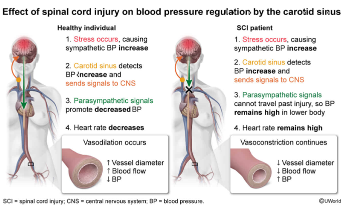
Figure 2 Calpain-I western blot (A) and quantified calpain-I expression levels (B) in each group

The carotid sinus is a structure in the neck that senses blood pressure changes and induces nervous system responses to maintain homeostasis. In patients with certain SCIs affecting upper regions of the spinal cord, blood pressure in the lower regions of the body cannot effectively return to normal levels following the stress response. The SCIs in these patients most likely:

- A. prevent the carotid sinus from relaying appropriate signals to somatic neurons throughout the body. [22%]
- B. promote negative feedback control of the sympathetic nervous system by the carotid sinus. [10%]
- ✔ C. block parasympathetic signals initiated by the carotid sinus from reaching some peripheral nerves. [58%]
- D. enhance positive feedback of the carotid sinus response on blood pressure levels. [8%]

Correct
57% answered correctly

Explanation:



The autonomic nervous system is made up of peripheral nerves that regulate **involuntary** bodily functions (eg, smooth muscle contraction in blood vessels, secretion of certain endocrine hormones).

The two divisions of the autonomic nervous system exhibit **antagonistic** effects as follows:

- The **parasympathetic division** promotes "rest-and-digest" responses such as *increased* digestive functions and *decreased* blood pressure (ie, via lower heart rate and vasodilation).

- The **sympathetic division** leads to "fight-or-flight" responses, including *increases* in blood pressure (eg, via vasoconstriction).

The autonomic nervous system may also interact with the endocrine system via neuroendocrine pathways. For example, the adrenal medulla is innervated by **sympathetic neurons**. As part of the stress response, these sympathetic neurons stimulate adrenal medullary cells to secrete norepinephrine and epinephrine, which can act to *increase* sympathetic responses (eg, blood pressure increases).

The question describes the carotid sinus, a structure in the neck that senses blood pressure changes and induces nervous system responses to maintain homeostasis. Patients with certain SCIs cannot effectively return blood pressure levels to normal in lower body regions following the stress response induced by the sympathetic division. In other words, the parasympathetic division is unable to *decrease* stress-induced blood pressure increases. Therefore, SCIs **most likely block parasympathetic signals** initiated by the **carotid sinus** from reaching some **peripheral nerves**, causing blood pressure to remain increased.

(Choice A) Somatic neurons regulate skeletal muscle (ie, voluntary) activities. However, blood pressure changes (eg, vasoconstriction/vasodilation) are involuntary and are regulated by the *autonomic* (*not* somatic) nervous system.

(Choice B) In negative feedback, the product of a biological pathway *inhibits* one or more of the steps in the same pathway. Therefore, negative feedback control of the sympathetic nervous system by the carotid sinus would likely *decrease* sympathetic responses, including *decreasing* blood pressure.

(Choice D) In positive feedback, a response is *amplified* because the product of a biological pathway *increases* production of the same product, moving an organism *further away* from the original state (ie, homeostasis). Because the carotid sinus response acts to *return* blood pressure to homeostatic levels, it likely does *not* regulate blood pressure using positive feedback.

Educational objective:
Within the autonomic nervous system, the sympathetic division promotes "fight-or-flight" responses (eg, increased heart rate and blood pressure) whereas the parasympathetic division promotes "rest-and-digest" responses (eg, increased digestion, decreased blood pressure).

Time spent:0 sec
Subject:
Biology

QID:403367
System:
Endocrine and Nervous Systems

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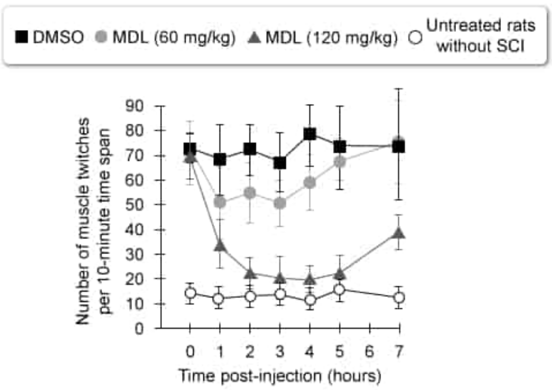

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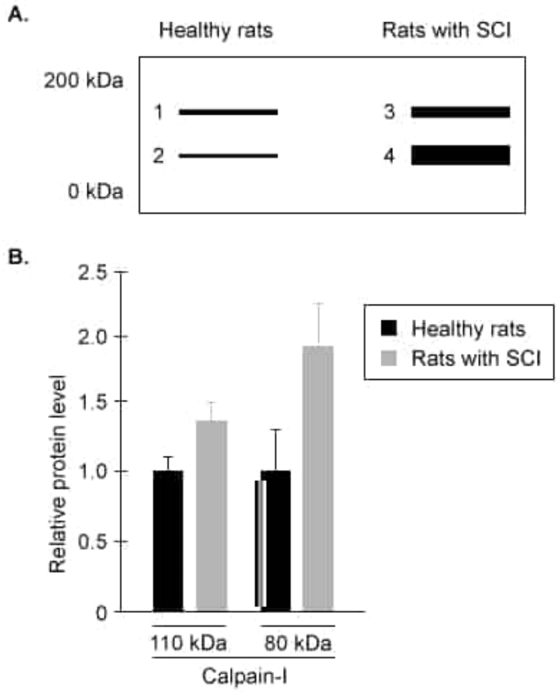


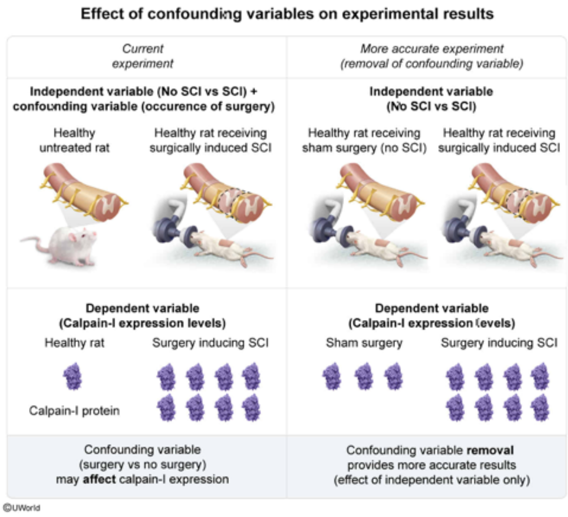
Figure 2 Calpain-I western blot (A) and quantified calpain-I expression levels (B) in each group

Which change to the experiment testing calpain-I levels in healthy rats and rats with SCI (Figure 2) would increase the accuracy of the experimental results?

- A. Increasing the number of healthy rats compared to rats with SCI included in the experiment [6%]
- B. Varying the locations of surgically induced SCIs among rats in the experimental group [20%]
- C. Including an additional group of rats in which the gene encoding calpain-I has been deleted [42%]
- D. Performing similar surgical procedures on the healthy rats that do not result in SCI [29%]

Correct
29% answered correctly

Explanation:



In a scientific experiment, the variable *manipulated* by researchers but *not* altered by other measured variables is called the **independent variable**. Researchers **measure** the response of a **dependent variable** to changes in the independent variable. Control variables should be held as constant as possible during an experiment to demonstrate that changes observed in the dependent variable were due *only* to changes in the independent variable (ie, results were *not* impacted by confounding variables).

In the experiment testing how SCI affects calpain-I expression levels in rats, two groups of rats were used: a healthy control group that was untreated and a group in which SCI was surgically induced at the same spinal cord location (the independent variable). Figure 2 shows that calpain-I expression (the dependent variable) *increased* in SCI rats compared to healthy rats.

However, in this experiment the control and SCI groups **differed** in that **only** the SCI group underwent a surgical procedure. Because stress from the surgical procedure itself could also affect calpain-I expression, this variable (ie, occurrence of a surgery) should be **controlled**. Therefore, to **increase the accuracy** of the experimental results, **healthy rats** should undergo **surgical procedures similar** to those performed on rats in the SCI group, with the only difference being that these surgeries would *not* result in SCIs (ie, **sham surgeries**).

(Choice A) Increasing the number of healthy rats compared to rats with SCI included in the experiment would cause the sizes of the control and experimental groups to be *unequal*, which would likely *decrease* the ability of the groups to be compared in a *fair* manner (ie, *decrease* the accuracy of experimental results).

(Choice B) Varying the locations of surgically induced SCIs among rats in the experimental group would introduce another variable (ie, a confounding variable) that could also impact calpain-I expression. Therefore, this option would likely decrease (*not* increase) the accuracy of these experimental results.

(Choice C) Rats in which the gene encoding calpain-I has been deleted would be *unable* to express calpain-I protein. Therefore, this additional group would *not* improve the accuracy of an experiment testing how SCI affects calpain-I *expression*.

Educational objective:
In an experiment, control variables should be held constant to better ensure that any changes to the dependent (measured) variable are only due to changes in the independent (manipulated) variable.

Time spent: 0 sec	**QID**:403368
Subject:	**System:**
Biology	Endocrine and Nervous Systems